

Public Hospital Utilization & Performance Analysis

This document outlines the key analytical areas and performance indicators derived from the public hospital utilization dataset. The analyses focus on understanding patient flow, service demand, hospital performance, and healthcare outcomes.

Key Analyses

- **Patient Admission Trends Over Time:** Analysis of how patient admissions vary across different time periods to identify patterns, peaks, and seasonal fluctuations in hospital usage.
- **State-wise Hospital Utilization:** Examination of patient admissions across states to understand geographic distribution of healthcare demand.
- **Hospital-level Utilization Comparison:** Evaluation of patient volume handled by each hospital to identify highly utilized and underutilized facilities.
- **Department-wise Patient Distribution:** Assessment of patient admissions across hospital departments to determine areas with the highest service demand.
- **Service Type Utilization Analysis:** Analysis of different healthcare services to understand which services are most frequently accessed by patients.
- **Gender-based Utilization Analysis:** Comparison of hospital usage between male and female patients to highlight demographic utilization patterns.
- **Discharge Performance Analysis:** Review of admitted versus discharged patients to evaluate hospital efficiency and patient throughput.
- **Mortality Analysis by Hospital and Department:** Examination of mortality counts and rates across hospitals or departments to assess patient outcomes and care quality.
- **Length of Stay Analysis:** Evaluation of average patient stay duration to understand efficiency in patient management and bed occupancy.
- **Geographic Distribution Analysis (Map Analysis):** Spatial analysis of patient admissions by state to visualize regional disparities and concentration of hospital utilization.

At-a-Glance Key Performance Indicators (KPIs)

- **Total Patients Admitted:** Represents the overall volume of patients served within the selected period.
- **Total Patients Discharged:** Indicates the total number of patients successfully discharged from hospitals.
- **Overall Mortality Rate:** Measures the proportion of patient deaths relative to total admissions, serving as a critical healthcare outcome indicator.
- **Average Length of Stay:** Reflects the average number of days patients spend in the hospital, indicating operational efficiency.

- Discharge Rate: Shows the proportion of admitted patients who were discharged, highlighting patient flow effectiveness.
- Most Utilized Department: Identifies the department with the highest patient volume.
- Top Performing Hospital by Admissions: Highlights the hospital with the highest number of admitted patients.
- State with Highest Hospital Utilization: Identifies the state recording the largest share of patient admissions.